Adam Driscoll

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Software engineer with experience in robotics, georegistration, localization, navigation, mapping, and 3D reconstruction

Work Experience

Software Engineer
Palantir | Python **Oct 2021–Jan 2026**

- Designed and built Unscented Kalman Filter that incorporates georegistration to estimate in-flight aerial vehicle global position in gps-denied settings
- Integrated the estimation algorithm into a simulation environment by utilizing Docker containers and the MAVlink protocol. Wrote a script to automatically generate and launch a flightpath in this environment
- Developed optical flow algorithm to enable real-time tracking of frames in full-motion videos by updating frame coordinates in-between executions of georegistration algorithm
- Adapted internal Python module capabilities to integrate georegistration tools into Palantir's Sensor Inference Platform, allowing it to output corrected geo-coordinates
- Implemented subpixel registration algorithm to improve the accuracy of our satellite georegistration process

Computer Vision Engineer II
Hivemapper | C++, Python **2019–2021**

- Designed, built, and maintained Open Telemetry Kit, an open-source Python tool to automatically extract drone camera and flight data to feed into the Hivemapper platform to improve 3D modeling and georegistration
- Developed algorithm to determine confidence in point cloud vertices and remove low-confidence points, improving final point cloud and mesh quality
- Enhanced approach to detecting and masking video overlays to video and extended this algorithm to apply during mesh texturing
- Investigated and validated various SLAM, VO, and depth estimation algorithms to improve accuracy and quality of maps generated from dashcam videos

Software Developer, Intern
Discovery Robotics | ROS, Gazebo **2018**

- Built simulation environment in Gazebo to enable rapid iteration on software algorithms
- Modeled flagship robot in simulation including accurate physics, motion model, and sensor representation

Operational Stability Engineer
Amazon Robotics | MySQL **2013-2017**

- Developed over 20 automation tools to replace manual task execution and reduce system failures
- Analyzed over 200 complex software issues and identified their root causes
- Created MySQL queries to collect data from 29 client facilities and presented this data in a UI using internally developed tools, allowing maintenance teams to efficiently analyze warehouse status

Education

- **M.Sc. Robotic Systems Development,** Carnegie Mellon University, Pittsburgh, PA **2018**
- **B.Sc. Robotics Engineering,** Worcester Polytechnic Institute, Worcester, MA **2012**

Projects

Groundsbot, Capstone
CMU, groundsbot.com | C++, ROS

- Designed, built, and programmed an autonomous field robot capable of mowing the rough grass at a golf course
- Created robust perception and localization subsystems by fusing data from Lidar, RTK GPS, IMU, and encoders
- Created GPS waypoint following and control algorithms used in the navigation subsystem of GroundsBot
- Achieved 98.6% coverage of input area, avoidance of 4/5 static obstacles and detection of 22/25 dynamic obstacles